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High-efficiency and visual-usability image encryption based on thumbnail preserving and chaotic system

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ABSTRACT

Privacy concerns may be caused after uploading images into image hosting platforms. For traditional image encryption, images with rich content and meaning are transformed into noise-like encrypted images without any visual information, which can protect privacy from infringement but at the expense of visual usability. Recently, thumbnail-preserving encryption (TPE) was proposed to balance privacy and usability by preserving the thumbnail unchanged after encrypting. On the other hand, a large number of pseudo-random functions with high computational complexity are utilized in the existing TPE schemes, which makes the encryption and decryption time too long. Motivated by this, we deeply analyze the characteristics of TPE and propose a scheme to organically combine TPE with chaotic systems, which reduces the encryption and decryption time exponentially besides ensuring security. The experiments show that the time cost is reduced dozens of times as well as relatively stable compared with the existing schemes. Meanwhile, encrypted images have high visual quality and can resist some commonly used attacks including differential attack and face detection and the balance between privacy and visual usability is achieved.

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1. Introduction

In the modern world, it has been becoming a part of daily life to take images with the rise of smart phones and thus the number of images has increased sharply. According to one report (Pantic, 2021), more over 1 trillion images were captured in 2020, despite the COVID-19 epidemic disrupting crucial occasions for taking images such as parties and travels. Meanwhile, one of the driving forces behind taking images is to post them on image hosting platforms such as social medias and cloud storage to record and share life due to people's social attributes (Pantic, 2021). For example, Instagram, one platform focusing on image sharing, has reached 1.16 billion users as of Oct. 2020 (KEMP, 2020).

As images may contain much information about privacy that may not be what users wish to expose (Li et al., 2018) such as health (Hua et al., 2018; Imtyaz Ahmed and Kannan, 2021) and

social relations (Sun et al., 2017), some privacy risks can be risen after uploading images into image hosting platforms (Amon et al., 2020). Furthermore, the effect of these images uploaded may deviate from the user's imagination, i.e., they are taken out of context (Nissenbaum, 2009). For instance, some humorous or embarrassing images may be crazily forwarded and spread like a virus (Amon et al., 2020), usually at the expense of the privacy of the image party (i.e., the user).

Meanwhile, image hosting platforms may not be reliable. For iCloud, it was broken by hackers in 2014 and the indecent images of some of the world's famous Hollywood stars were leaked (Marwick, 2017); and recently, it announced that it would cooperate with the U.S. government to scan the images stored in it (Greenberg, 2021), which caused huge privacy disputes, although it claimed to protect children. For Facebook, it was found to have illegally transmitted user data containing images to the Cambridge company (Cadwalladr and Graham-Harrison, 2018) and the company acknowledged that images are automatically scanned when they are sent on Messenger (Davies, 2018). In addition, one artificial intelligence (AI) company, Clearview AI, was exposed to collect massive facial images from image hosting platforms without the user's consent and thus created a large-scale face detection model,

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which tries to accurately connect the identity with hosting platform accounts (Meden et al., 2021).

Traditional image encryption (Liu et al., 2021; Xian and Wang, 2021; Wu et al., 2021; Maazouz et al., 2022), i.e., an original image with rich information is transformed into a noise-like encrypted image (Wen et al., 2020), is an effective technology to deal with image privacy. On the other hand, it gives up the usability, which may bring many inconveniences for users. This not only makes it impossible for others to distinguish between images but also for users to do the same, which means that users cannot select the desired image based on their own preferences before downloading and decrypting images. In a nutshell, visual usability, which is also important to users, is abandoned for the sake of image privacy. Zhao et al. pointed out that privacy protection schemes should not be implemented at the expense of usability to be successful among users (Zhao et al., 2021).

There seems to be a hostile relationship between usability and privacy, but fortunately, some user surveys pointed out that the both can be carried out at the same time. Denning et al. proposed a password authentication system using image degradation and they carried out user experiments to verify the privacy and usability (Denning et al., 2011). In the experiments, users were required to browser some images of common objects in advance and then the degradation version of these images was taken to users, which were required to provide the label most suitable for the content. The experimental results showed that users could recognize the image degradation version normally and stably, while an adversary who has not seen the image could only make random attempts. Similarly, EvoPass was verified through the image degradation (Yu et al., 2017). User experiments demonstrated that users who have browsed the image could effectively recognize the degradation image that the resolution is poor whereas others were not able to. In a words, if the encrypted image contains the degraded version of the image, it can protect privacy besides obtaining visual usability.

Thumbnail-preserving encryption (TPE), in which the ideal or approximate thumbnail is considered as a kind of image degradation version, was proposed to balance privacy and usability by preserving the degradation version of the original image in 2015 (Wright et al., 2015). Specifically, the thumbnail of the image encrypted by the TPE scheme is entirely or almost same to the thumbnail of the original image, resulting that accurate visual con-

tents are eliminated and rough visual ones are preserved in TPE images. Therefore, this brings advantages over the original image or image encrypted by traditional encryption uploaded to the platform as shown in Fig. 1. In short, users who have previously seen the image can obtain the usability by previewing TPE images but others who have not seen it cannot learn the accurate privacy information as shown in Fig. 1(c). Compared with original images as shown in Fig. 1(a), TPE images can protect the image privacy; compared with images by traditional image encryption as shown in Fig. 1(b), users can gain visual usability from previewing TPE images.

So far, existing TPE schemes can be classified as ideal TPE (Zhao et al., 2021; Wright et al., 2015; Tajik et al., 2019) or approximate TPE (Marohn et al., 2017; Zhang et al., 2022) according to whether the thumbnail preserved in the encrypted image is ideal or approximate in comparison with the original thumbnail. For approximate TPE schemes, HF-TPE (Zhang et al., 2022) is the stage of the art at present. Although the thumbnail of the encrypted image in HF-TPE is almost consistent with the original thumbnail and the difference between the decrypted image and corresponding original one cannot be seen visually, this approximation or difference is nevertheless improper in specific scenarios requiring accuracy like medical ones. For ideal TPE schemes, Tajik et al. (2019) proposed a novel framework to construct ideal TPE, which attracted some follow-up researches for TPE (Zhao et al., 2021; Zhang et al., 2022). However, this framework needs to call a large number of pseudo-random functions (PRF) with high computational complexity, e.g., the number of PRF called to encrypt an image with size $x \times y$ is greater than $\frac{xy}{2}$, which brings a great cost of time.

Chaos theory is widely utilized to the encryption processing (Singh et al., 2021; Hua et al., 2021) since it has many features, e.g., unpredictability, ergodicity, and initial value sensitivity (Frank et al., 2014), similar to the requirements of image encryption (Lan et al., 2018). Existing chaos systems can be divided into two categories (Hua et al., 2021; Lan et al., 2018; Kaur et al., 2021), one-dimensional (1D) system and multi-dimensional (MD) one, based on the dimension of the phase space. For the 1D system, it may cause security risks when used for image encryption (Hu et al., 2017) since its chaotic signal can be expected due to its simple structure and thus the processing of encryption can be easily predicted. For the MD system, it has at less two parameters, resulting in the sophisticated structure and difficult to the predict

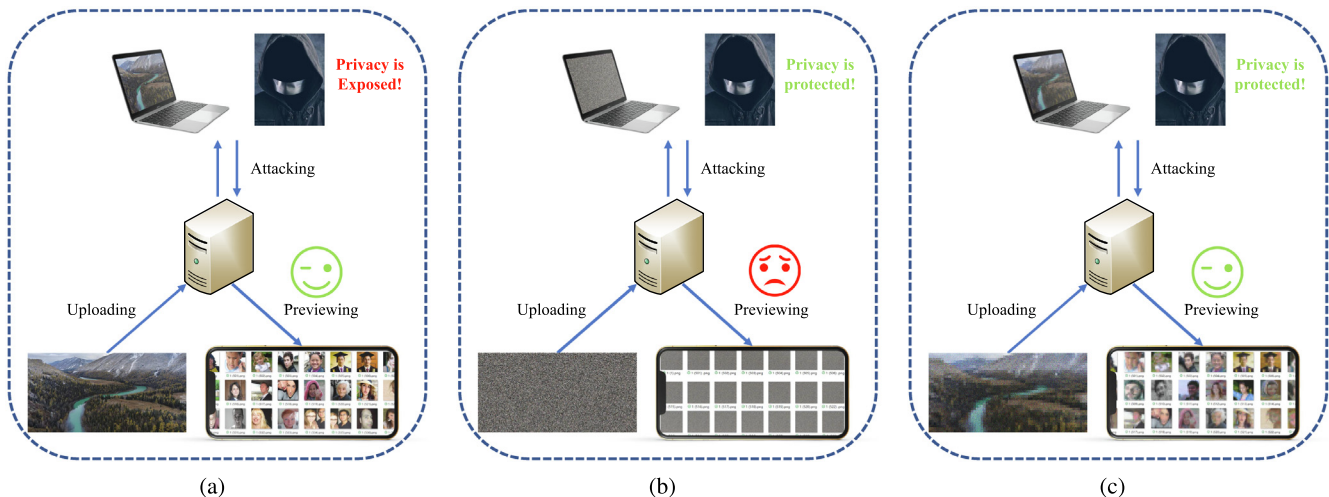


Fig. 1. The effects after images processed by different means uploading to the image hosting platform. (a) An original image is uploaded. Although users can enjoy the visual usability, privacy attackers can obtain the precise privacy information in the image; (b) An image encrypted by the traditional image encryption is uploaded. Privacy is protected from privacy attackers but users cannot preview these images; (c) A TPE image is uploaded. Images can be used for users by previewing and privacy attackers cannot learn the precise privacy contents.

behavior but at a high time cost (Hua et al., 2021). Two-dimensional (2D) chaotic systems can have the complex and unpredictable behavior besides the low time cost, leading to be widely utilized in image encryption (Hua et al., 2021).

Motivated by this, we present the visual-usability image encryption based on thumbnail preserving and chaotic system. We deeply analyze the internal relationship between TPE and 2D chaotic system and then combine both organically. In a words, the chaotic system is utilized to generate random numbers and permutation map, which are used for substitution and permutation (SP), respectively. As a result, the classical architecture of cryptography is achieved, i.e., confusion-diffusion, and the time cost is greatly reduced compared with existing TPE schemes called with many PRFs. The overall comparison of our scheme with the representative TPE schemes (containing virtually all schemes) is depicted in Table 1, showing that no other scheme outperforms the proposed one in any metric, i.e., the proposed scheme is optimal as a whole.

Contribution: The main contributions of this paper are drawn as follows:

- We deeply analyze the characteristics of TPE and chaotic systems, and then they are combined to exponentially reduce the time cost whether the encryption or decryption. As far as we know, it is the first scheme to utilize the chaotic system to construct the TPE and only 1 s is taken for encrypting a color images with size 512×512 , which is the fastest TPE scheme using SP framework.
- We propose threat models and design goals for the scenario of the image hosting platform. Guided by this, the proposed TPE scheme can be good applicable to the platform and users can get a great experience, i.e., user-friendly. Meanwhile, it is the first TPE scheme to achieve all goals.
- We do some experiments to demonstrate the performance. Their results show that the encryption time is shorter compared with other TPE schemes and the visual quality is great whether the encrypted image or decrypted one. Meanwhile, some secure experiments denote that the scheme can protect privacy in addition to having certain robustness, e.g., the encrypted image can block face detection algorithms.

The rest of the paper is organized as follows. Section 2 briefly refers some works that try to balance privacy and usability. Section 3 is some preliminaries utilized in the proposed scheme while threat models and goals are introduced in Section 4. The details of constructing the scheme are described in Section 5. Experiments and analyses are listed in Section 6. Discussions are presented in Section 7 and Section 8 concludes this work.

2. Related work

Some researchers begin to pay attention to the contradiction between image privacy and usability and put forward the following schemes to alleviate the incompatibility with them.

2.1. Privacy prediction

Privacy prediction schemes (Orekondu et al., 2017; Tonge and Caragea, 2020; Tonge and Caragea, 2019) do not directly process the image but scan it to determine whether there is privacy information and then remind users to make a further decision. Orekondu et al. proposed a scheme to predict the privacy by combining image contents with user preferences, and some experiments was done to prove that the scheme can successfully predict specific privacy and even better than the user's own judgment (Orekondu et al., 2017). Subsequently, Tonge et al. designed a scheme utilizing convolutional neural networks (CNNs) to improve the prediction performance by about 10% (Tonge and Caragea, 2020). Although these schemes seem to can protect privacy by predicting, there are some problems. First, these schemes are based on image contents and assume that privacy is tied to objects, i.e., an image is determined to be private as long as a private object defined in advance appears in the image. However, privacy is more likely to be context-based rather than content-based as mentioned in Vishwamitra et al. (2021). Second, they require users to have a large enough data set with privacy labels, which may be a burden for users. In addition, simply informing users about the privacy of images without processing is insufficient in the hosting platform scenario.

2.2. Image filter

Image filter, e.g., mosaic, pixelation, and Gaussian blur, is a common method to balance privacy and usability in daily life. Google Street View utilizes the blurring to deal with private contents such as faces and car numbers (Frome et al., 2009). In TV news interviews, the parties' faces are often protected by mosaics to protect the privacy (Du et al., 2014). Ilia et al. designed a fine-grained scheme that visitors obtain the original image or the image with the blurred face based on the rank of them (Ilia et al., 2015). Zhou et al. recently proposed the first scheme to blur irrelevant faces, i.e., bystander faces, in multimedia data (Zhou and Pun, 2021). These schemes achieve good results in the privacy protection of specific objects like faces, but some problems remain with them. First, the detection of specific objects in the image is not totally effective and some objects that should be detected may be missed. Second, the existing schemes can only effectively deal with a few simple objects such as faces and license plates, which are closely related to privacy but still insufficient for fully protecting privacy (Nodari et al., 2012).

2.3. Object replacement

Object replacement can protect image privacy by replacing sensitive objects with insensitive ones while the content of other parts has not changed resulting in the visual usability. Face de-identification (Li and Lin, 2019) is a classic privacy protection technology. It protects privacy by replacing real faces with false ones which may be synthetic or extracted from face data sets. Face

Table 1

The overall comparison of our scheme with the existing TPE schemes. Scheme 1 to Scheme 6 are Wright (Wright et al., 2015), TPE-LSB (Marohn et al., 2017), DPRE (Marohn et al., 2017), Tajik (Tajik et al., 2019), TPE2 (Zhao et al., 2021), and HF-TPE (Zhang et al., 2022), respectively.

Item	Scheme						Proposed
	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Scheme 5	Scheme 6	
Quality of Encrypted Image	Ideal	No Ideal	No Ideal	Ideal	Ideal	No Ideal	Ideal
Quality of Decrypted Image	Ideal	No Ideal	No Ideal	Ideal	Ideal	No Ideal	Ideal
Decryption Success Rate	100%	100%	≠ 100%	100%	100%	100%	100%
SP framework	No	No	No	Yes	Yes	No	Yes
Encrypted time cost	Quick	Quick	Quick	Slow	Very Slow	Slow	Quick
Decrypted time cost	Quick	Quick	Quick	Slow	Very Slow	Slow	Quick

attribute privacy (Morales et al., 2021) is a more fine-grained technology compared with de-identification to balance privacy and usability. Some attributes in the face rather than the whole face are regarded as privacy information and eliminated, but other attributes are not changed resulting in usability. However, these schemes are irreversible and thus unsuitable for some scenarios of the image hosting platform such as cloud storage. In addition, there are too many unchanged places in the processed image and thus too much information is exposed, which may affect privacy.

2.4. Region of interest encryption

Region of interest (ROI) encryption divides the image into privacy parts and public ones according to the privacy of image contents. For the former, encryption schemes are utilized to encrypt it for preventing privacy disclosure. For the latter, it is not processed since it is regarded as a non sensitive without privacy information. Zhang et al. proposed an encryption scheme considering the contour feature in the image as ROI (Zhang et al., 2013). He et al. proposed a novel ROI encryption scheme and the authors claimed that it does not scarify the image usability besides protecting privacy (He et al., 2016). However, as mentioned in Section 2.1, how to divide the privacy area in the image? Although the image segmentation algorithm (Minaee et al., 2021; Ouahabi and Taleb-Ahmed, 2021) has made a breakthrough with the development of CNNs, it is an objective and universal technology. For privacy, it is a kind of subjective and unstructured information, which means that privacy is difficult to be correctly separated by the segmentation algorithm.

3. Preliminary

3.1. Thumbnail-preserving encryption

The generation method of the thumbnail considered by TPE is shown in Fig. 2, which can be divided into three steps as follows:

- An image M with size $x \times y$ is inputted and the size of the thumbnail block is determined as $b \times b$. Note that x and y can be divisible by b ;
- M is divided into thumbnail block sets $\{B_{1,1}, \dots, B_{x/b, y/b}\}$;
- The average of all pixel values in each block $B_{i,j}$ ($1 \leq i \leq \frac{x}{b}, 1 \leq j \leq \frac{y}{b}$) is calculated, which is one pixel value in the thumbnail.

The above generation method means that if the sum of pixel values in each thumbnail block preserves unchanged after encrypt-

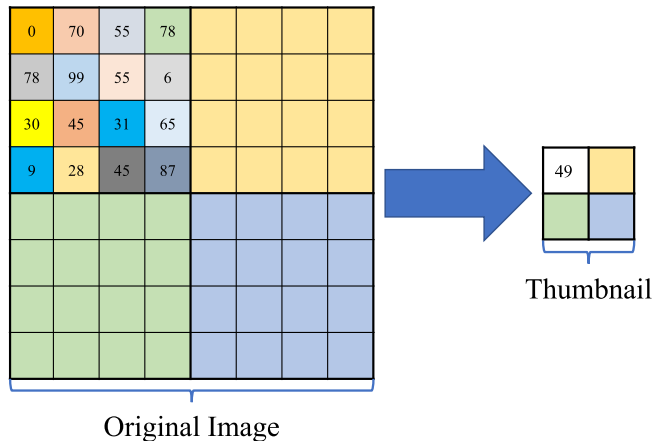


Fig. 2. An example of generating the thumbnail.

ing, then the thumbnail will be preserved the same as the original one. Therefore, the problem of the thumbnail preserving after encrypting can be simplified to preserve the sum of pixel values in each block unchanged, i.e., sum-preserving encryption (SPE).

Bellare et al. (2009) proposed the format-preserving encryption scheme is suitable for constructing SPE and the steps are as follows:

- A numerical vector $\vec{v} \in \mathcal{M}$ is used as an input. Note that $\mathcal{M}$ denotes the message space and $\mathcal{M} = (\mathbb{Z}_{d+1})^n$;
- The sum s of $\vec{v} = \{v_1, \dots, v_n\}$ is calculated, i.e., $s = \sum_{i=1}^n v_i$;
- All vectors whose sum is s and belongs to $\mathcal{M}$ are enumerated, which form a vector set $\Phi_{sum}^{-1}(s)$;
- Each vector in $\Phi_{sum}^{-1}(s)$ is given an index number, which is unique in $\Phi_{sum}^{-1}(s)$;
- Two functions, $\text{rank}(\cdot)$ and $\text{rank}_s^{-1}(\cdot)$, are designed, which convert a vector to the corresponding index number and an index number to the corresponding vector, respectively;
- $\vec{v}$ is inputted into $\text{rank}(\cdot)$ and thus an index number r_o is obtained;
- r_o is encrypted to r_e which is a legitimate index number in $\Phi_{sum}^{-1}(s)$, i.e., r_e can correspond a vector in $\Phi_{sum}^{-1}(s)$;
- r_e is inputted into $\text{rank}_s^{-1}(\cdot)$ and thus an encrypted vector $\vec{c}$ is obtained. By the way, $\vec{c} \in \mathcal{M}$ and $s = \sum_{i=1}^n c_i$.

3.2. Two-dimensional logistic-sine mapping

Although the 1D chaotic system cannot realize the complex chaotic behavior, it has simple structure and low implementation cost. The chaotic behavior of MD system is sophisticated and difficult to predict, but it has the problems of complex structure and high cost. Motivated by this, Hua et al. (2021) proposed a 2D chaotic mapping by using two nonlinear 1D chaotic systems, i.e., 2D logistic-sine mapping (2D-LSM), as follows:

$$\begin{cases} x_{i+1} = \cos(4 \times a \times x_i \times (1 - x_i) + b \times \sin(\pi \times y_i) + 1) \\ y_{i+1} = \cos(4 \times a \times y_i \times (1 - y_i) + b \times \sin(\pi \times x_i) + 1) \end{cases} \quad (1)$$

where a and b are two parameters and they can be any value. The authors made many experiments to prove that the system has better chaotic behavior than other 2D ones, strong randomness, and wide out range besides continuous.

4. Threat model and goal

4.1. Threat model

In this work, two privacy threats are considered, i.e., internal threat and external threat, as follows:

Internal threat: Users cannot control the access and behavior of the hosting platform to images after uploading. Meanwhile, both platform providers and malicious employees have sufficient motivation to abuse this access right to browser images for benefits. In addition, many hosting platforms utilize multi-backup mechanism. Even if users opt to delete the images from the platform, the backup data is not be deleted, which means that images will have a privacy threat for a long time after uploading no matter what users do.

External threat: Privacy threateners from the outside may monitor and eavesdrop the communication between clients and servers of the hosting platform on insecure channels. Meanwhile, they may also utilize abnormal means such as user enumeration attacks and brute force password cracking to obtain the correct user account along with password and thus enter the hosting platform to steal images. Furthermore, they may also utilize some vulnerabilities

to bypass the password authentication mechanism and directly obtain the images stored on the server.

4.2. Goal

Visual observability: The image encrypted by the proposed scheme should preserve enough visual contents related to the original image and thus legitimate users can obtain usability through direct observation with the naked eye.

Protecting privacy: The encrypted image should not expose fine visual contents of the original image to protect privacy.

Balance tunability: Everyone may have a distinctive opinion, tolerance, and balance point for privacy and visual usability since the two are subjective concepts. Therefore, **users should easily tune the balance between the two according to their own wishes.**

Universality: As mentioned above, privacy and usability are both subjective concepts. Whether ordinary users or algorithms along with manual or automatic, it is difficult to distinguish privacy and usability. Meanwhile, privacy is nuanced and closely related to the context, which means that every part of the original image may reveal privacy. **Therefore, the scheme should be a universal process for the whole image rather than distinguish each part in an image to separately process.**

Lossless reversibility: The encrypted image **should be losslessly decrypted** to the original image to apply some accurate scenes. Meanwhile, if the encrypted image cannot be losslessly decrypted, each encryption and decryption will accumulate losses and finally cause a great damage to the image quality.

Low time cost: The encryption and decryption times for a user-friendly scheme should be low. It is not acceptable for many users to spend tens of seconds encrypting an image before uploading. Meanwhile, a long encryption time often means a large amount of computing, i.e., needing many device resources, which also affects of the use of the scheme by users.

Low size expansion: The size of the encrypted image should not be too large as the encrypted image is transmitted between the client and the server. The large file size can result in inefficient transfers, low storage efficiency, and lengthy transfer times.

5. Construction

5.1. Overview

The images considered in the proposed are as follows:

- The image is composed of one or more channels;
- The lossless compression algorithm is utilized in the image format, e.g., **PNG** format and **BMP** format;
- The image has a **unique identifier T** , which means that no two images' identifiers are the same. The metadata of the image can act as the identifier as do the existing TPE schemes (Zhao et al., 2021; Tajik et al., 2019; Zhang et al., 2022);
- Each channel of the image is a 2D numerical matrix, in which each pixel value belongs to $\mathbb{Z}_{d+1}$. For many image formats, **$d = 255$** , which is a default in the paper.

In addition, users are **assumed** like other TPE schemes (Zhao et al., 2021; Tajik et al., 2019; Zhang et al., 2022) as follows:

- Users have a proven secure **PRF**. Any probabilistic polynomial-time (PPT) algorithm, i.e., adversary, cannot distinguish the pseudo random number generated by the PRF from the real random number $l \xleftarrow{\$} \{0, 1\}^n$ (Goldreich et al., 1986), where $\$$ means random and unbiased selector;

- Users have an adequately sophisticated key $k \xleftarrow{\$} \{0, 1\}^\lambda$, where λ denotes a security parameter large enough. **The key has been fully cryptanalyzed to prove that can resist the attack of the PPT adversary;**
- In this work, the secure PRF and key k are combined to form the function $G_k(\cdot)$, which is used for deriving random numbers. Researches have proven (Luby and Rackoff, 1988) that a secure PRF can be applied to successfully construct a pseudo-random number generator that **can prevent the chosen-plaintext attack**. Meanwhile, the probability of the same output by the generator $G_k(\cdot)$ can be ignored provided the inputs are different.

5.2. Chaotic matrix

In this work, Eq. (1) is utilized to generate two chaotic sequences, in which initial parameters a, b, x_0 , and y_0 are set and inputted by users. Eq. (1) is iterated $w = q + x \times y$ times, in which $q = q_1 + q_2$, while x and y denote the size of the inputted image. Chaotic systems have transient effects (Falcioni et al., 2005), i.e., the sequences generated at the beginning of the chaotic system, **the first q_1 bits, exhibit the characteristics of irregularity, instability, and non chaos.** These sequences are not suitable for image encryption due to the transient effect and thus they need to be discarded. **q_2 presents a random numbers generated by PRF and key, i.e., $G_k(\cdot)$.** It is used to ensure the security of the generated sequence and the mechanism for generating it will be described later.

Therefore, the first q bits in the generated sequence are discarded and then two chaotic sequences with length $x \times y$ are obtained, i.e., $s_x = \{x_{q+1}, \dots, x_{q+x \times y}\}$ and $s_y = \{y_{q+1}, \dots, y_{q+x \times y}\}$. **Then, the two 1D sequences are adjusted to 2D matrices with size $x \times y$, which are called S and P , respectively.**

5.3. Encryption

A color image often comprises multiple channels and the thumbnail of the image will preserve unchanged after encrypting **if the sum of element values of the thumbnail block in each channel preserves unchanged.** Therefore, it is divided into multiple channel matrices and separately processed in each channel after an original image M with size $x \times y$ is inputted. The overview framework of the proposed scheme is shown in Fig. 3.

In this work, the size of the thumbnail block is $b \times b$ and each channel matrix of the image M is divided into blocks $M_b[i, j, c]$, where i and j are the index of the block in the 2D matrix ($1 \leq i \leq \frac{x}{b}, 1 \leq j \leq \frac{y}{b}$). c represents the number of the channel and an image often has three channels and thus $c = 0, 1, 2$ by default.

As Tajik et al. said that the time complexity of directly processing the all elements in the whole $M_b[i, j, c]$ by SPE is too large, but it is simple to deal with two elements as a time (Tajik et al., 2019). Therefore, the elements in $M_b[i, j, c]$ are firstly **grouped into element pairs $\vec{g} = (a, b)$** and then $\vec{g}$ is processed by SPE as follows.

First, the function $G_k(\cdot)$ is called to generate a random number $q_2 = G_k(T||c||e)$, where $||$ means **connector that connects two strings** and **e denotes the number of encryption rounds that will be mentioned later.** In short, T and c are unique for each image and the channel of the image, respectively, and meanwhile, e is different in each round of encryption. Therefore, the input $T||c||e$ is different in each channel and round of encrypting each image and thus q_2 generated by $G_k(T||c||e)$ is unparalleled. Then, chaotic matrices with size $x \times y$, i.e., S and P , are generated as shown in Section 5.2. S and P are utilized for substitution encryption and permutation encryption of the image, respectively.

For the **substitution encryption**, all elements in S do the following:

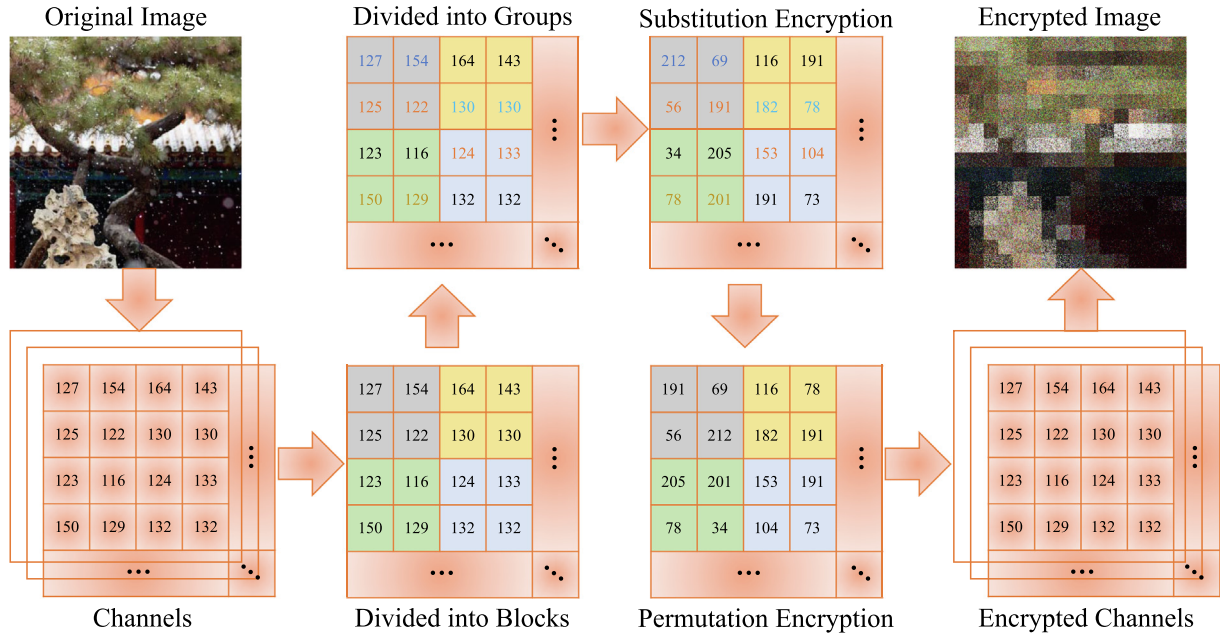


Fig. 3. The overview of the proposed scheme.

$$S_1[ei, ej] = \begin{cases} S[ei, ej], & S[ei, ej] \geq 0 \\ -S[ei, ej], & \text{otherwise} \end{cases} \quad (2)$$

where $S[ei, ej]$ represents the (ei, ej) -th element in the matrix S . A matrix S_1 is obtained after all the elements in S are calculated by the Eq. (2). Subsequently, S_1 is also divided into blocks $S_{1b}[i, j, c]$ according to the size of the thumbnail block and then all elements in $S_{1b}[i, j, c]$ are grouped into element pairs $\vec{h} = (sa, sb)$.

Taking the element group $\vec{g} = (a, b)$ in thumbnail block $M_b[i, j, c]$ as an example, the processing of substitution encryption is the following:

- The sum of $\vec{g} = (a, b)$ is calculated, i.e., $s = a + b$;
- The sum of $\vec{h} = (sa, sb)$ corresponding to $\vec{g}$ is also calculated, which is multiplied by a large enough number and finally rounded, i.e.,

$$|t = (sa + sb) \times \tau + \frac{1}{2}|, \quad (3)$$

where τ denotes a large enough number and $\lfloor \cdot \rfloor$ means round down. By the way, a number is rounded down after adding $\frac{1}{2}$ is equivalent to round it;

- The function $\text{rank}(\cdot)$ with two elements is defined as follows:

$$\text{rank}(a, b) = \begin{cases} a, & s \leq d \\ a - s + d, & \text{otherwise} \end{cases} \quad (4)$$

- The function $\text{rank}_s^{-1}(\cdot)$ with two elements is defined as follows:

$$\text{rank}_s^{-1}(r) = \begin{cases} (r, s - r), & s \leq d \\ (s - d + r, d - r), & \text{otherwise} \end{cases} \quad (5)$$

- The index number r_o of $\vec{g}$ is extracted by function $\text{rank}(\cdot)$;
- r_o is encrypted as follows:

$$r_e = r_o + t \bmod |\Phi_{sum}^{-1}(s)|, \quad (6)$$

where $|\Phi_{sum}^{-1}(s)|$ means the number of vectors in $\Phi_{sum}^{-1}(s)$ and the symbol of the mod stands for modulo operators. The equation of $|\Phi_{sum}^{-1}(s)|$ is as follows (Zhang et al., 2022):

$$|\Phi_{sum}^{-1}(s)| = \begin{cases} s + 1, & s \leq d \\ 2 \times d - s + 1, & \text{otherwise} \end{cases} \quad (7)$$

- The function $\text{rank}_s^{-1}(\cdot)$ is utilized to convert r_e to the encrypted group $\vec{c}$ that the sum of the element value is s .

The example of image substitution encryption is shown in Fig. 4. It can be seen that the sum of element values in each thumbnail block does not change after substituting encryption.

Then the permutation encryption process is carried out as follows:

- The chaotic matrix P is divided into blocks $P_b[i, j, c]$ with size $b \times b$;
- The elements is numbered in each block $P_b[i, j, c]$ according to the size of element values. A matrix P_1 is formed after all blocks $P_b[i, j, c]$ are numbered;
- According to P_1 , each thumbnail block in the image is separately permuted encryption.

The example of image permutation encryption is shown in Fig. 5. Simply put, after the elements in every $P_b[i, j, c]$ are numbered in the order of size of the values, the elements of the thumbnail block in the substituted image are permuted in the order of the number. The encrypted image has the same thumbnail as the original image and the thumbnail will not be changed even if the encrypted image is encrypted again as input. It is called as one encryption round as shown in Fig. 6 after performing one substitution alongside with permutation, which can be many rounds for security when the image is encrypted.

5.4. Decryption

The operation of the image decryption is simply introduced since it is similar to but reverse to the encryption operation, which is as follows:

- The chaotic matrices S and P are generated according to parameters, and then they are processed to produce t and P_1 ;
- Each thumbnail block in encrypted image is performed permutation decryption according to P_1 ;



Fig. 4. The example of the substitution encryption and the sizes of the image and thumbnail block are 4×4 and 2×2 , respectively: (a) the process of generating t from chaotic matrix S and $\tau = 10000$; (b) the process of image substitution encryption.

- Each two elements in each thumbnail block are used to form one group;
- The index number r_e of each group is extracted by the function $rank(\cdot)$;
- r_e is **decrypted** to obtain the original index r_o as follows:

$$r_o = r_e - t \bmod |\Phi_{sum}^{-1}(s)|; \quad (8)$$

- r_o is inputted into $rank_s^{-1}(\cdot)$ to output the original group;
- The image is decrypted once all groups have done so.

The above decryption process considers that the encrypted image has only one channel and only one encryption round is made. For the encrypted image with multiple channels and many encryption rounds, the decryption can be simply completed by repeating the above process.

6. Experiment

In this section, the performance is verified from various aspects, e.g., time cost, image quality, size expansion, histogram analysis, etc. By the way, the experiments have the most types and are comprehensive compared with the existing TPE schemes, which not

only covers representative TPE experiments but also many experiments that are utilized in TPE for the first time, fully demonstrating the advantages of the proposed scheme. General experimental information is shown below:

Experimental platform: All experiments are performed on one computer, and its main parameters are i7-8700 CPU @ 3.2 GHz, 16 GB RAM, and Window 10.

Image dataset: The first 1000 images in Helen dataset¹ (Le et al., 2012) are used in experiments. Meanwhile, all images are converted to PNG format and resized with 512×512 .

In addition, image encryption and decryption as well as other experiments are carried out using Matlab, except that the face detection experiment is employed with Python.

6.1. Time cost

In this part, comparison schemes are the scheme proposed in Tajik et al. (2019) (called Tajik) and HF-TPE (Zhang et al., 2022) since all of them are two elements as an unit to perform the substitution encryption at the one time like the proposed scheme. Tables 2–4 present general evaluations of the encryption time of

¹ <http://www.ifp.illinois.edu/vuongle2/helen/>

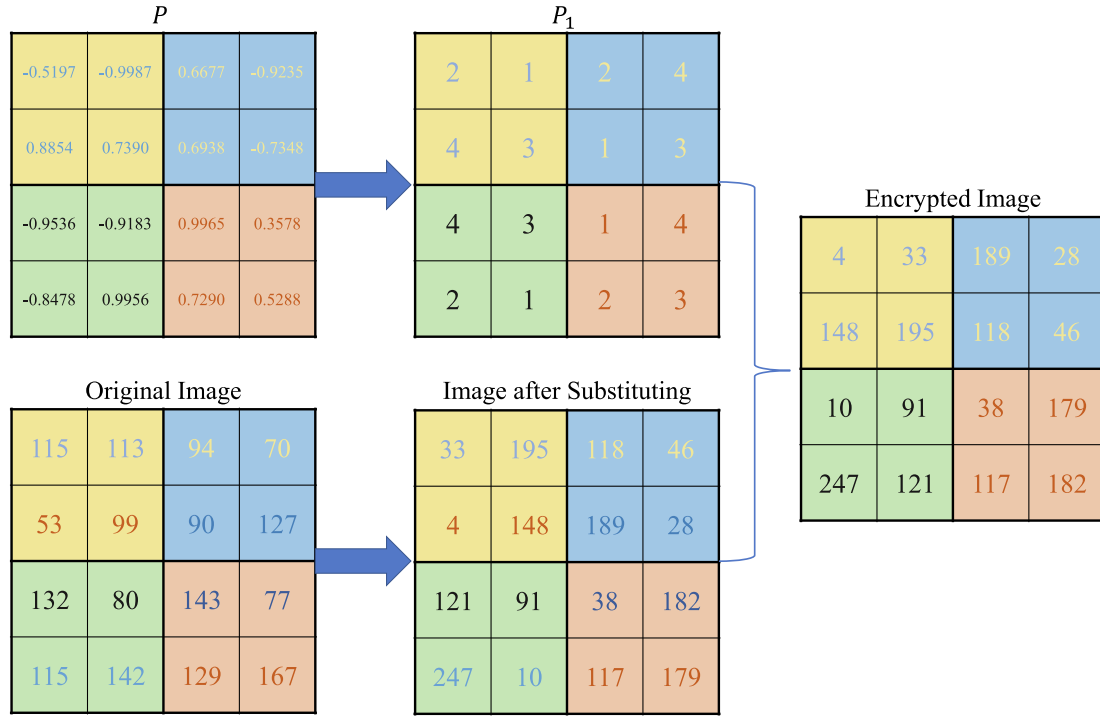


Fig. 5. The example of permutation encryption, in which the sizes of the image and thumbnail are 4×4 and 2×2 , respectively.

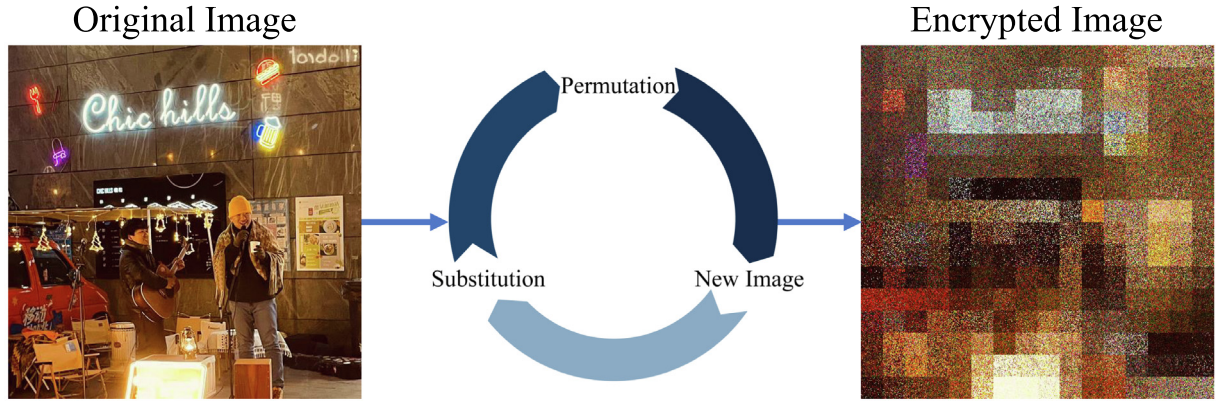


Fig. 6. The encryption round of the image.

the proposed scheme, Tajik, and HF-TPE, respectively, which seen that the average time of the proposed scheme is about **60 times and 70 times** faster than the other two, respectively. Meanwhile, the encryption time is also stable, which can be seen from the standard deviation and the difference between the maximum time and minimum time.

Further, the encrypted time of the three schemes is recorded in each image of the dataset under different sizes of the thumbnail block and then the fluctuation between the encrypted time of each image and the average time is calculated as follow:

$$\varpi = \left| \frac{\gamma - \bar{\gamma}}{\bar{\gamma}} \right|, \quad (9)$$

where ϖ means fluctuation, γ presents an encrypted time of one image and $\bar{\gamma}$ denotes the average of the encrypted time. Fig. 7 shows the time fluctuation in different block sizes and the pink line in the middle represents the situation where the fluctuation is zero. The point above the line expresses the case that its fluctuation is greater

than the average time; the point below indicates the opposite case. For the proposed scheme, most of the points are tightly around the pink line and their fluctuation values do not exceed than 0.05, which represents the stability of the encrypted time. By the way, the decryption time of the three schemes are not listed since they are roughly the same as the encryption time.

6.2. Image quality

The effects of image quality of existing TPE schemes are shown in Table 1, in which the term 'ideal' in the quality of encrypted image means that the thumbnail of the encrypted image is the same as that of the original one, and in the quality of decrypted image means that the decrypted image is the same as the original one. The goal for the quality of the encrypted image is to preserve the thumbnail consistent with the thumbnail of the original image as much as possible, which presents that the quality of encrypted image in the proposed scheme has reached the theoretical optimal value, i.e., 'ideal'. Similarly, the effect of decrypted image is also the

Table 2

Time cost of the proposed scheme (units in seconds).

Item	Scheme			
	The proposed scheme			
	8 × 8	16 × 16	32 × 32	64 × 64
Average Time	1.5308	1.4038	1.3840	1.3788
Standard Deviation	0.0015	0.0009	0.0010	0.0011
Maximum Time	1.7415	1.6635	1.5186	1.5950
Minimum Time	1.4480	1.3405	1.3273	1.3166

Table 3

Time cost of Tajik (units in seconds).

Item	Scheme			
	Tajik			
	8 × 8	16 × 16	32 × 32	64 × 64
Average Time	92.6479	92.1738	93.9273	93.3785
Standard Deviation	6.5845	6.7535	5.9441	5.9920
Maximum Time	104.7847	109.1916	108.8878	101.5601
Minimum Time	89.0296	89.0634	90.0390	89.9943

Table 4

Time cost of HF-TPE (units in seconds).

Item	Scheme			
	HF-TPE			
	8 × 8	16 × 16	32 × 32	64 × 64
Average Time	110.3894	107.9098	110.2784	107.4650
Standard Deviation	3.0385	2.8809	3.0943	3.9875
Maximum Time	119.3679	115.7901	120.8373	127.3896
Minimum Time	94.5851	94.8455	94.4825	92.9793

theoretical optimal value. Therefore, the proposed scheme is not weaker than any existing one in terms of image quality.

Meanwhile, experiments are operated in this part and the visual effects of images encrypted firstly are listed as shown in Fig. 8, in which five daily images with size 512×512 are encrypted under different block sizes. It can be seen that the smaller the block size is, the better the visual usability is and the worse the privacy is; conversely, the larger the block size is, the worse the visual usability is and the better the privacy is, meaning that the balance between privacy and usability can be adjusted by the block size.

Further, objective indicators are utilized to evaluate image quality, i.e., structural similarity (SSIM) and peak signal to noise ratio (PSNR), which are commonly used to evaluate image (Ouahabi, 2013). The quality of the encrypted image and decrypted one of all ideal-TPE schemes (Wright et al., 2015; Tajik et al., 2019; Zhao et al., 2021) is consistent with the proposed scheme and thus they are no longer listed. Meanwhile, DPPE is excluded since the encrypted image may be failed to decrypt in the scheme. Therefore, approximate-TPE schemes, i.e., HF-TPE (Zhang et al., 2022) and TPE-LSB (Marohn et al., 2017), are selected to be comparison schemes. By the way, TPE-LSB approximates the thumbnail by flipping the lowest bits of each pixel. The image quality can be controlled by controlling the parameter u of the number of flipping the lowest bits. For simplicity, different parameters u are regarded as different schemes, which is denoted by TPE (u bit).

In addition, SSIM prefers to evaluate the image quality from the whole image compared with PSNR, which is line with the characteristics of the quality of the TPE image. Meanwhile, PSNR is more inclined to evaluate the noise in the image compared with SSIM, which conforms to evaluate the decrypted image. In fact, whether the encrypted image or the decrypted one, the proposed scheme achieved the best results in the both indicators compared with

other schemes. For the evaluation of the encrypted image, TPE images are converted to thumbnails with size 150×150 since the size of the thumbnail on platforms is fixed regardless of the size of the thumbnail block selected by users. Fig. 9(a) shows the results of this experiment, denoting that the image encrypted by the proposed scheme has the best thumbnail effect compared with others. For the evaluation of the decrypted image, the original image is directly compared with the decrypted image and the results is shown in Fig. 9(b). Note that the proposed scheme is not listed in Fig. 9(b) since the decrypted image is the completely same as the original one, causing the PSNR value to be infinite. As can be seen from Fig. 9, the image quality is the best compared with others for both encrypted images and decrypted ones.

6.3. Image size expansion

As mentioned in Section 4.2, the image size is important in terms of image transfer, storage, and time. In this part, the size expansion is calculated by the ratio of the sizes of the processed image to the original image and '1' means no expansion. The results of the encrypted image expansion are illustrated in Fig. 10(a), showing that the value of the proposed scheme in this indicator is less than 2.2 times. Compared with the traditional image encryption scheme that the size of the encrypted image is expanded hundreds (Paillier, 1999) or even thousands of times (Brakerski et al., 2012), the size expansion of the encryption of the proposed scheme can be ignored and it is friendly for users. The results of the decrypted image expansion are depicted in Fig. 10(a), and the value of the proposed scheme is 1 in this index, indicating no size expansion. For users, it is more important the decrypted image with no size expansion than the encrypted image with some expansion since the former is stored at the local end

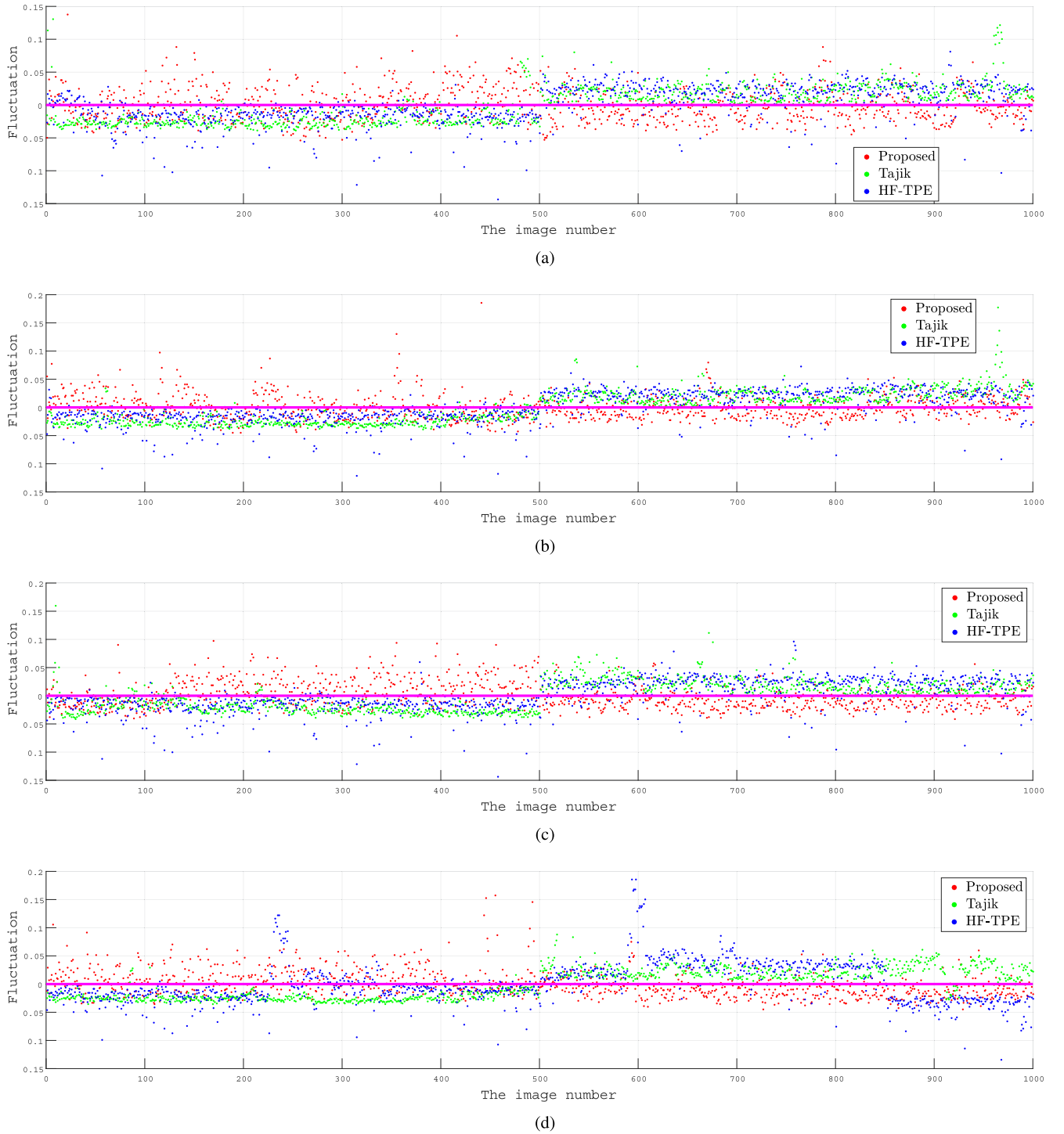


Fig. 7. The encrypted time fluctuation of each image under different sizes of the thumbnail block: (a) 8×8 ; (b) 16×16 ; (c) 32×32 ; and (d) 64×64 .

with limited storage space while the latter is saved at the platform (Zhao et al., 2021).

6.4. Face detection

The ability of the TPE image to resist the detection algorithm is tested in this part while the face detection algorithm is selected since it is one of the most commonly used detection algorithms and the face is highly related to the privacy (Adjabi et al., 2021; Adjabi et al., 2020). In this subsection, the face detection API of

Face++² is selected for this experiment while TPE2 (Zhao et al., 2021) and HF-TPE (Zhang et al., 2022) are the comparative schemes since their image quality is similar to that of the proposed scheme.

Fig. 11 denotes the results in this experiment and shows that the ability of the proposed scheme to resist face detection algorithm is similar to that of the other two schemes. Meanwhile, the empirical value of anti detection algorithm is 32×32 obtained from the experiment when the image size is 512×512 .

² <https://www.faceplusplus.com.cn/>

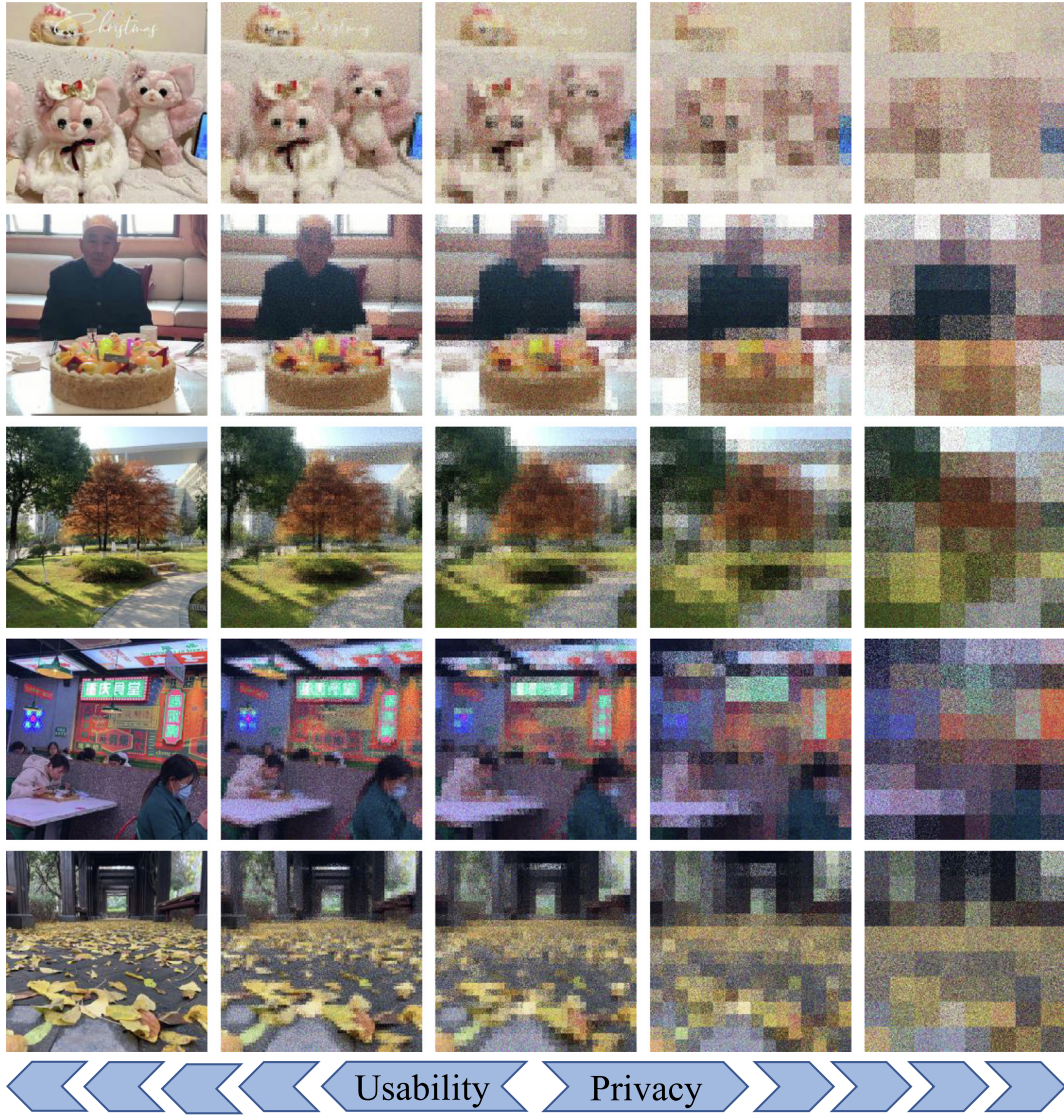


Fig. 8. The effect of images encrypted by the scheme under different sizes of the thumbnail block. From left to right are 0×0 (i.e., original image), 8×8 , 16×16 , 32×32 , and 64×64 , respectively.

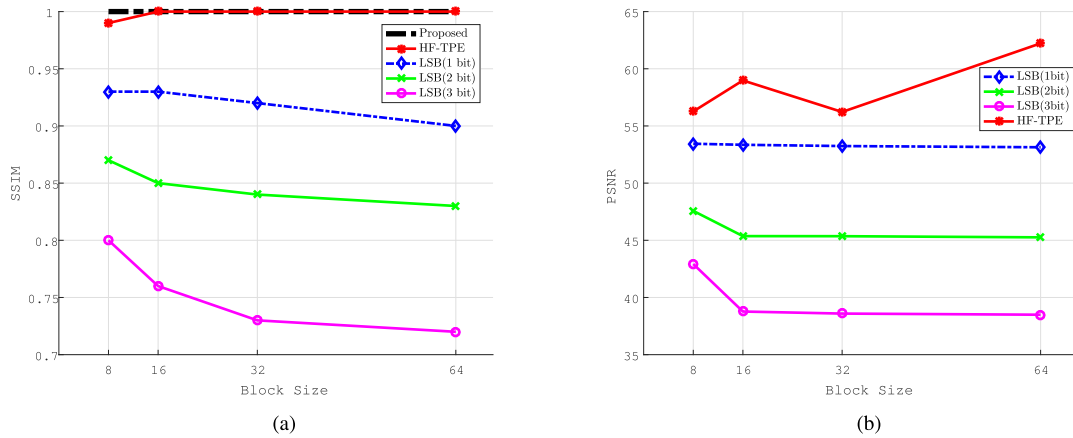


Fig. 9. Quality evaluation of encrypted and decrypted images: (a) the encrypted image evaluated by SSIM; (b) the decrypted image evaluated by PSNR. Note that the proposed scheme is not included in Fig. 9(b) since its PSNR value is infinite.

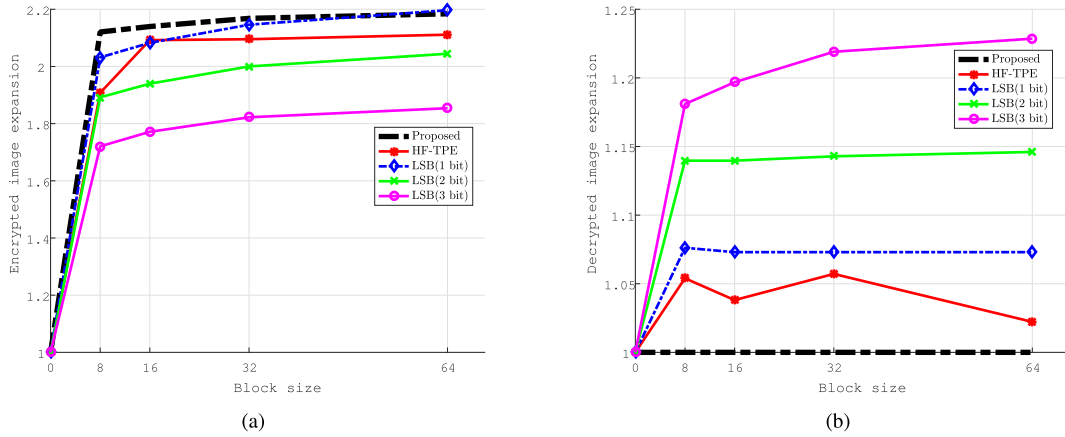


Fig. 10. The evaluation of the expansion of the image size: (a) the encrypted image; (b) the decrypted image.

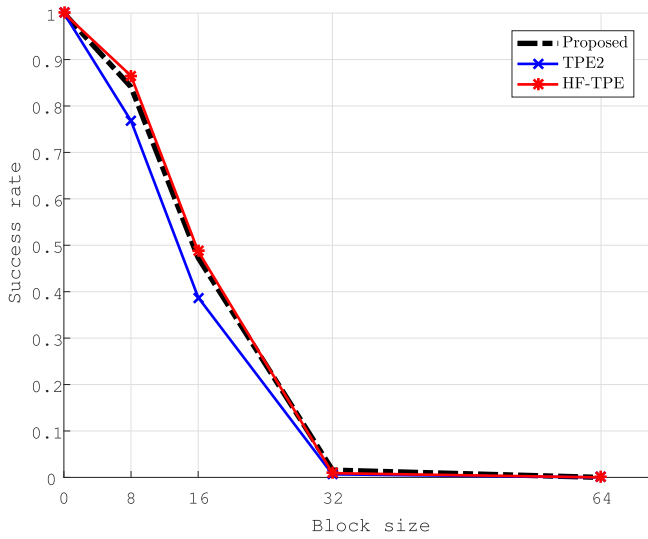


Fig. 11. The success rate of the face detection.

6.5. Histogram analysis

The histogram analysis of encrypted images is an important analysis method for the security of an image encryption algorithm. For a secure TPE algorithm, the histogram on each channel should be stabilized after a certain number of rounds of encryption.

In this part, some images are randomly selected to encrypt and observed their histogram after encrypting some rounds. Fig. 12 is an example of one of the images and their histograms on each channel show in Fig. 13, denoting that the histogram of the encrypted channel tends to stable after a certain number of rounds. Note that all images in this experiment have similar effects.

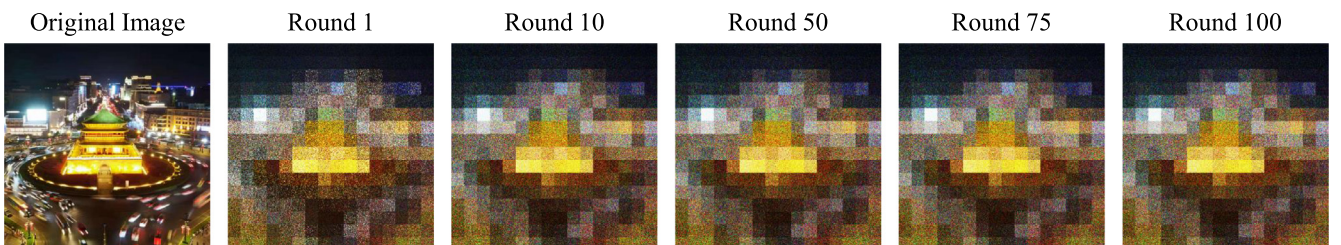


Fig. 12. The example image of different encrypted rounds.

6.6. Key sensitivity

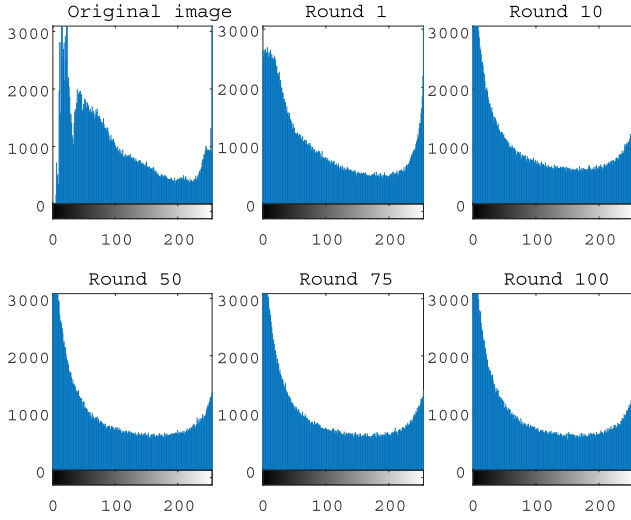
One of the important index of security in an image encryption scheme is key sensitivity, which means the degree of change of the encrypted image after a little change of key. In the proposed scheme, the key is used as an input in a security PRF to generate a random number q_2 that is employed to produce the chaotic matrix. As mentioned in Section 5.1, a security PRF means that it has been proved to be sensitive to input, large enough output space, output indistinguishable from randomly selected turn random numbers, and able to resist chosen-plaintext attack (Luby and Rackoff, 1988). Therefore, key sensitivity of the proposed scheme is enough in theory.

Furthermore, an image is separately encrypted by two keys with a difference of 1, i.e., 10000 and 10001, and other parameters are the same. Then, the subtraction is operated between the two encrypted images and the absolute value of the result is calculated, which is shown in Fig. 15. The result image, i.e., the rightmost end of Fig. 15, demonstrates that the degree of change between the image encrypted by different keys is huge even if the key is only a little different.

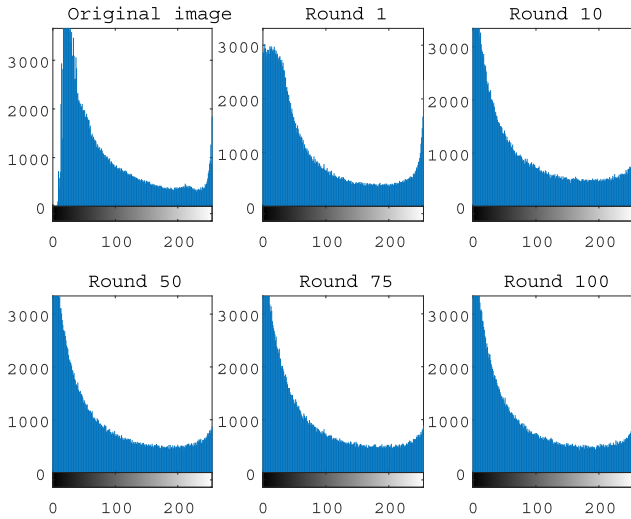
Meanwhile, the experiment of decrypting the encrypted with the wrong key is done as shown in Fig. 16, presenting that it can not be reversed to the original image as long as the key is changed a little, and the wrong decrypted image cannot be distinguished from the encrypted image visually. Thus, the proposed scheme is sensitive enough to the key.

6.7. Differential attack

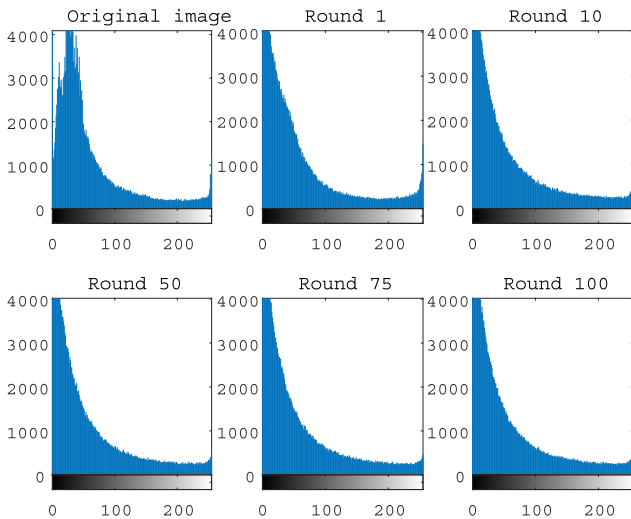
The evaluation of the differential attack is one of comment tests about the security of the scheme security. In this attack, **two images with only small difference are selected to be encrypted and then their results are compared.** For a good encrypted



(a)



(b)



(c)

Fig. 13. Histograms of the channels of image encrypted by different rounds: (a) R; (b) G; (c) B.

algorithm, the change of the encrypted image should be obvious even if the original image has only slight changes.

The number of pixel change rate (NPCR) (Wu et al., 2011) is one commonly used indicator to evaluate the ability of encryption algorithms to resist differential attack. In this part, one pixel value in each thumbnail block is changed since TPE is carried out in each block and then two encrypted images, DA_1 and DA_2 , are obtained. NPCR is calculated as follows:

$$NPCR(DA_1, DA_2) = \frac{\sum_{i=1}^x \sum_{j=1}^y D(i, j)}{H}, \quad (10)$$

where x and y denote the size of the inputted image, H means the number of pixels in each channel, and $D(i, j)$ is as follows:

$$D(i, j) = \begin{cases} 0, & DA_1(i, j) = DA_2(i, j) \\ 1, & \text{otherwise} \end{cases}. \quad (11)$$

The encryption scheme is considered to have a good ability to defense the differential attack if the NPCR value is greater than 0.99 according to the scheme (Wu et al., 2011). In the experiment, three images, Airplane, Baboon, and Peppers, are served as test examples as shown in Fig. 17 and are encrypted for many rounds. Table 5 shows the test results, which demonstrates that the proposed scheme can achieve a good performance in the aspect of the NPCR value and owns a good ability to resist the differential attack.

6.8. Chosen-plaintext attack

In the mode of the chosen-plaintext attack, the attacker can access to the encryption machine and can construct the encrypted image from any original one. Meanwhile, the attacker analyzes a certain number of original and encrypted images, which tries to deduce the relationship between them, and then decrypts the encrypted image without keys.

The proposed scheme can resist this attack with strong ability and the reason is as follows: **[1]** The chaotic system (Hua et al., 2021) used in the scheme has been demonstrated by a large number of experiments that it can produce chaotic sequences with high randomness and is extremely sensitive to initial parameters. **[2]** The secure PRF, which has been proved to have some features, i.e., sensitive to inputs, large enough output space, output indistinguishable from randomly selected true random numbers, and resisting chosen-plaintext attack (Luby and Rackoff, 1988), is introduced to generate random numbers to intercept the chaotic sequence. **[3]** Each image has a unique parameter T compared to any other image. T is inputted into the secure PRF with input sensitivity and thus the value generated by each call to PRF is unique. **[4]** The proposed scheme adopts the substitution-permutation architecture which is often used in many popular encryption algorithms such as DES and AES, and can be executed many times. Therefore, even small changes can diffuse throughout the full image.

Furthermore, the same experimental method as in the reference (Hua et al., 2021), i.e., encrypting an image twice with the same key, is used to visually show the ability of the proposed scheme to resist this attack and the test results are shown in Fig. 18. C1 and C2 in Fig. 18(a) are the two encrypted images, and the right-most end in Fig. 18(a) shows the difference between C1 and C2. Meanwhile, the histogram of each channel of $|C1 - C2|$ is depicted in Fig. 18(b). It can be seen that the two encrypted image are completely different, which demonstrates that the proposed scheme can get new and different encrypted images in each encryption.

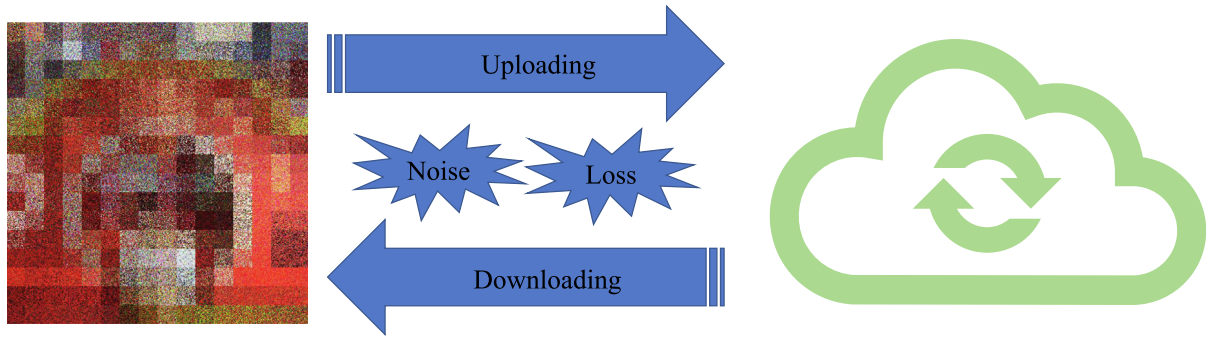


Fig. 14. The example of data loss and adding noise during image transmission.

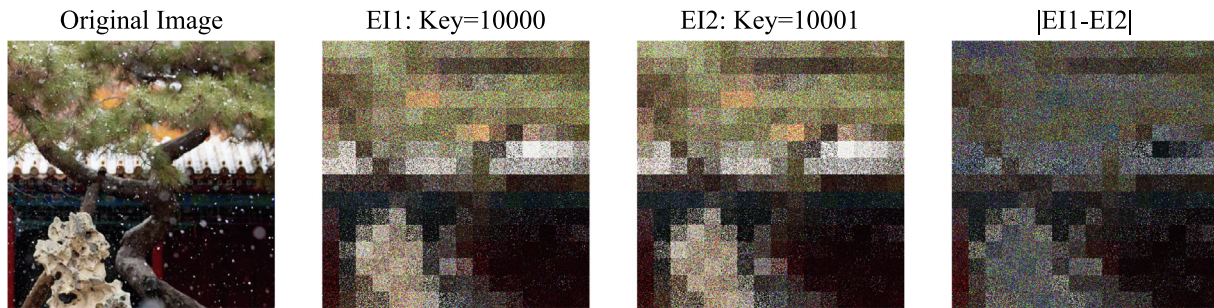


Fig. 15. The example of key sensitivity. From left to right: original image, EI1: encrypted image and key = 10000, EI2: encrypted image and key = 10001, and $|EI1 - EI2|$.

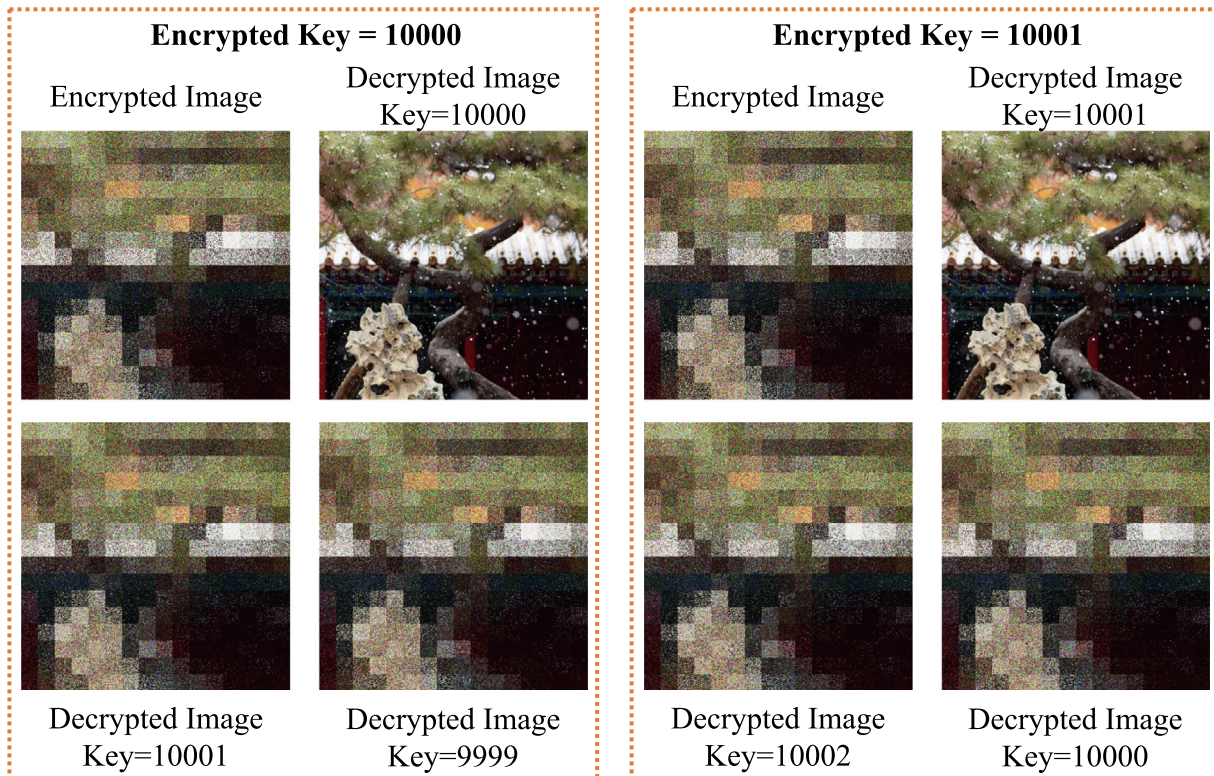
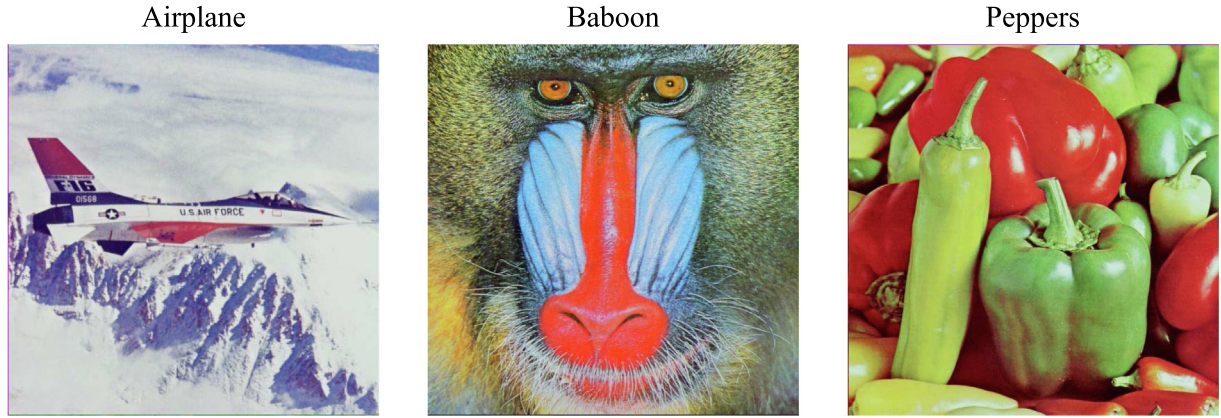
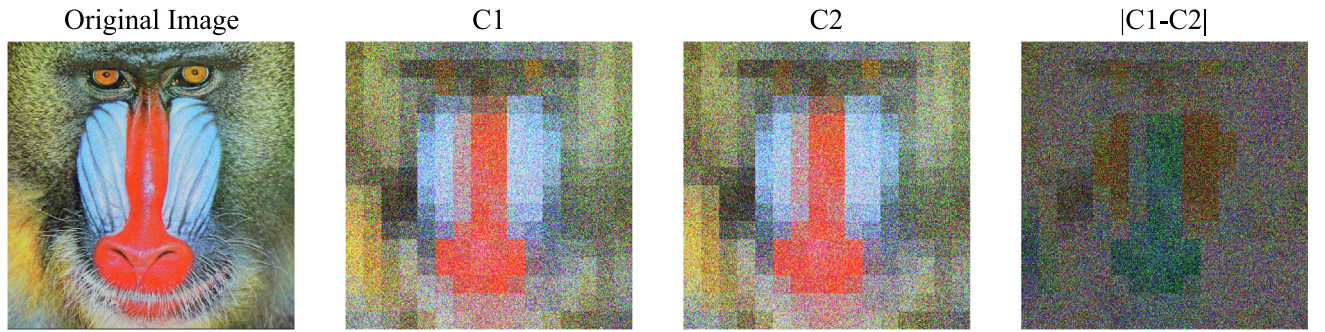
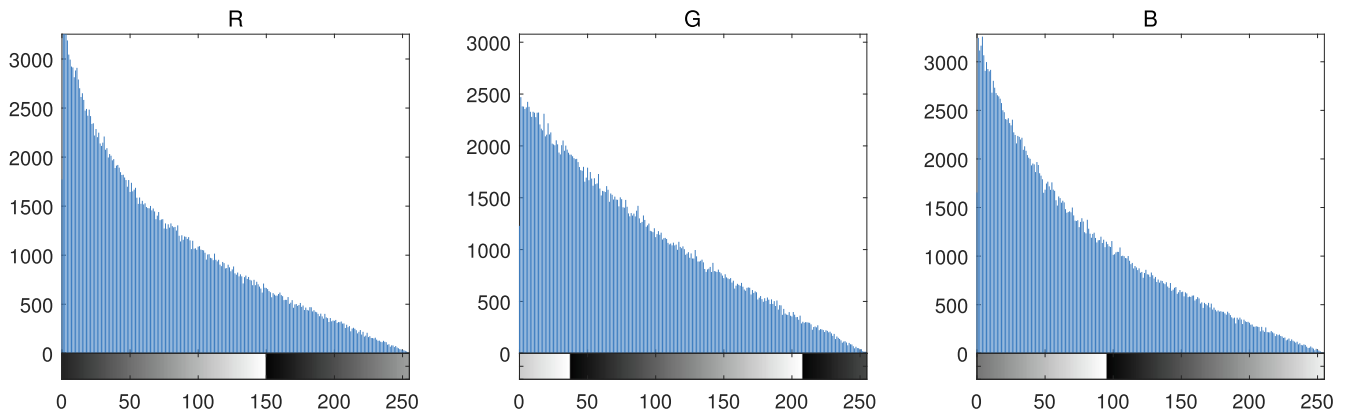


Fig. 16. The example of decryption separately employing the correct key and the key with small error.

Table 5

The NPCR value for different images and encryption rounds.

Image	Round							
	25	50	75	100	125	150	175	200
Airplane	0.99219	0.99222	0.99246	0.99235	0.99214	0.99237	0.99233	0.99226
Baboon	0.99396	0.99433	0.99424	0.99414	0.99412	0.99413	0.99420	0.99408
Peppers	0.99161	0.99171	0.99185	0.99159	0.99181	0.99172	0.99178	0.99169

**Fig. 17.** The text image of differential attack.**(a)****(b)****Fig. 18.** The example of the ability of resisting chosen-plaintext attack. (a) Images from left and right: original image, the first encrypted image C1, the second encrypted image C2, and difference image between C1 and C2, i.e., $|C1 - C2|$; (b) The histogram of $|C1 - C2|$.

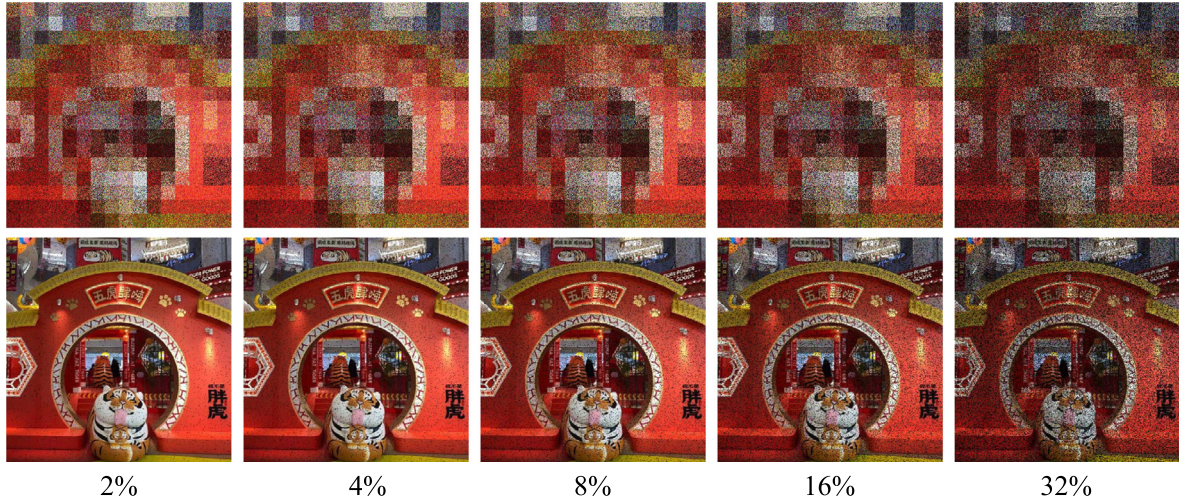


Fig. 19. The example image of data loss.

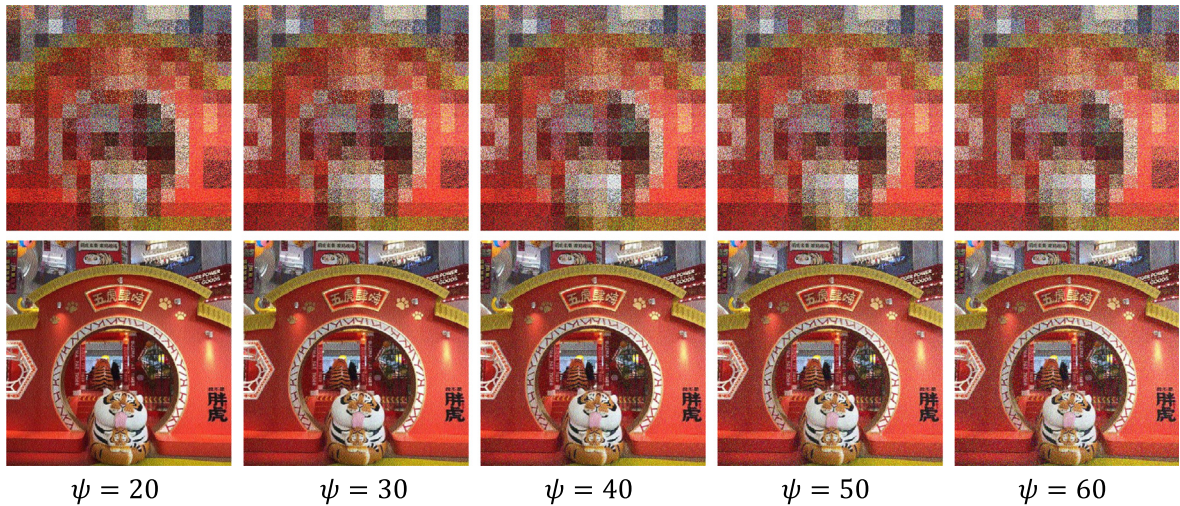


Fig. 20. The example image of noise attack.

The proposed scheme has a good ability to resist generally employed attacks containing chosen-plaintext attack according to this feature.

6.9. Robustness

Data loss and noise may be added into images during transmission as shown in 14. For a good encryption algorithm, it should be robust to these attacks to a certain extent, i.e., both encrypted images and decrypted ones after being attacked should preserve enough visual information that users can learn the usability by previewing.

For data loss, it is simulated by randomly losing some pixels in the TPE image as shown in Fig. 19, in which 2%, 4%, 8%, 16%, and 32% pixels are lost. It can be seen that the visual meaning of both the attacked TPE image and decrypted image is almost the same as that of non attacked image, except that the image is darkened.

For noise attack, it is simulated by randomly adding white Gaussian noise G with the average 0 and the variance 1 as follows:

$$I' = I \times \psi \times G + I, \quad (12)$$

where I' means the image added the noise, I denotes the normal image, and ψ presents the noise intensity. Fig. 20 is the example

of adding the noise of different intensity into the image, showing that the visual content of the TPE image and decrypted one is hardly affected.

7. Discussion

In Section 4.2, seven goals are proposed to protect the image privacy as well as preserve the visual usability. The following checks to see if the proposed scheme achieves these goals.

Visual observability: In the abstract, the thumbnail of the original image is preserved in the corresponding image encrypted by the proposed scheme and the visual content of the thumbnail can be directly observed by previewing the encrypted image as shown in Figs. 8 and 12. Therefore, the goal of the visual observability is achieved.

Protecting privacy: The visual content of the original image smaller than the block size, which is fine details and considered as the privacy content, is erased in the encrypted image. By the way, the privacy content is defined and depicted this way in all existing TPE schemes (Zhao et al., 2021; Wright et al., 2015; Marohn et al., 2017; Tajik et al., 2019; Zhang et al., 2022). Therefore, the goal of protecting privacy is acquired.

Balance tunability: In the scheme, the large block size, the better the privacy; conversely, the better the usability, which is shown in Fig. 8. In other words, the balance between privacy and usability can be tuned by the block size that is entirely user-determined. Thus, the goal of the balance tunability is achieved.

Universality: Each part of the image is treated the same by the proposed scheme without any exceptions and users do not need to distinguish the privacy sensitivity, which is difficult for both people and algorithms as depicted in Section 4.2, of each part. Therefore, the goal of the universality is achieved.

Lossless reversibility: In short, there are two encryption processes, substitution and permutation, in the scheme as shown in Fig. 6. For the substitution encryption, two pixels are extracted the index number by function $\text{rank}(\cdot)$ and then it is encrypted/decrypted by Eq. (6)/(8). Lastly, the encrypted/decrypted group is generated by function $\text{rank}_s^{-1}(\cdot)$, which is obviously lossless. Alternatively, the same matrix P is used in the permutation encryption/decryption of the same image, which is also lossless. Therefore, the goal of the lossless reversibility is finished.

Low time cost: The time cost of the proposed scheme is reduced by dozens of times compared with other similar schemes (i.e., substitution and permutation encryption based on two pixels) and the cost is close to 1 s for the first time as demonstrated in Tables 2–4. Accordingly, the goal of the low time cost is reached.

Low size expansion: In the scheme, the content of the encrypted image has a certain degree of redundancy since it preserves the rough visual content for the usability, leading that it can be compressed to a certain extent. Meanwhile, the experiment of the size expansion is demonstrated that the average size expansion of the encrypted image is only about 2 times, which is much lower than some image encryption algorithms such as homomorphic encryption. Thus, the goal of the low size expansion is achieved.

As a result, the all goals stated in Section 4.2 are achieved in the proposed scheme and no other TPE scheme has achieved all of the above objectives so far.

Meanwhile, the image format with lossless compression algorithm is considered in the scheme due to the goal of the lossless reversibility. In other words, the image format using lossy compression algorithm, e.g., JPEG, is excluded from the scheme, which can avoid some problems. For example, each encryption and decryption will bring some quality losses for lossy formats. Although this loss may be small, the image after undergoing multiple encryption and decryption may change significantly compared with the original one, i.e., this loss is accumulated.

8. Conclusion

In this paper, a high-efficiency and visual-usability image encryption scheme based on thumbnail and chaotic system is proposed. It exponentially reduces the time cost of encryption and decryption in addition to ensure the security by employing the confusion-diffusion characteristics of chaotic systems. Meanwhile, the scheme can have both privacy and usability, i.e., balance them, and the balance can be tuned by adjusting the size of the thumbnail block. Furthermore, experiment results show that images encrypted by the proposed scheme have highly visual quality and the privacy can be protected besides robustness. In comparison with other TPE schemes, the experiments in this paper have the most types and are comprehensive, covering not only representative experiments appearing in other TPE schemes but also many experiments that are utilized in TPE for the first time, fully demonstrating the advantages of the proposed scheme.

In short, the scheme achieves seven preset goals and may bring some advantages towards science and life. First, it is the first scheme that combines TPE and chaotic system as far as we known,

and it has better effects than other TPE schemes. Second, the substantial reduction of time cost of TPE makes some progress towards practical applications. Third, it may alleviate the contradiction between image privacy and visual usability after uploading images if the scheme can be applied. Fourth, the scheme is more feasible for users compared with other privacy protection solutions employing anonymization and image transformation, since it does not require users to have intricate knowledge like training models and professional equipment like GPU.

In the future work, some possible directions may further improve the existing TPE schemes. First, it may further help to enhance the effect of TPE schemes if the concept of privacy is clarified. The definition of privacy content in all TPE schemes is visual information smaller than thumbnail blocks at present and many similar schemes about image privacy protection also apply some subjective interpretation for privacy. If privacy can be explained clearly and even mathematically, it may make qualitative changes to TPE, i.e., the amount of information leaked in the encrypted image can be easily given quantitatively. However, privacy is a subjective concept as has been mentioned above, which may be difficult to clarify. Second, compatibility with JPEG format may make the scheme more applicable, but due to the above JPEG problems, the goal of lossless reversibility is contradictory to this format. Therefore, it may be considered to abandon this goal in the future to research how to achieve compatibility.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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